THE LONG RUN, EXTENDED

1,000,000 rounds. *The interval tightens.*

Ten times the sample. The same six decks, the same one-unit chart, and the same first 100,000 rounds.



NET RESULT

-3,672.00 units

W–L–P	OBSERVED EDGE	95% WIDTH
433,685–476,011–90,304	0.367%	0.451 pts

6 decks · S17 · 3:2 · Double any 2 · DAS · 3 resplits

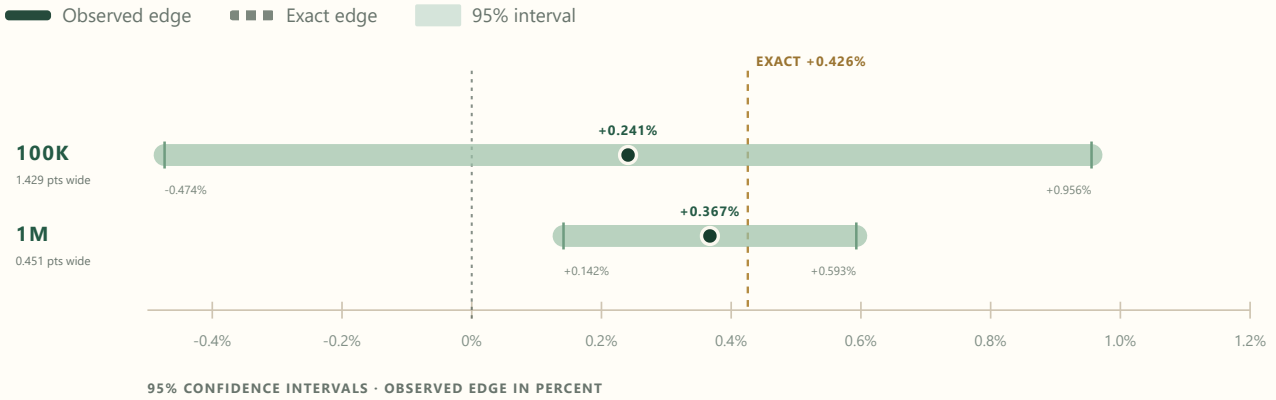
SEED 8675309 · 75% PENETRATION · COMMIT 6e0fc28

THE PRECISION STORY

The error bar *collapses.*

The first 100,000 rounds of this run are the 100k report. Extending to 1,000,000 rounds moves the observed edge from **0.241%** to **0.367%** and shrinks the 95% interval from **1.429 points wide** to **0.451**.

OBSERVED EDGE	EXACT EDGE	95% INTERVAL	WIDTH REDUCTION
0.367%	0.426%	0.142%	3.17×
1,000,000 rounds	composition-aware solver	to 0.593%	68.4% narrower



95% CONFIDENCE INTERVALS · OBSERVED EDGE IN PERCENT

The 100k interval included break-even. The 1M interval does not. The exact 0.426% edge sits inside both.

Measure	100,000	1,000,000	Change
Observed edge	0.241%	0.367%	+0.126 pts
95% interval	-0.474% to 0.956%	0.142% to 0.593%	3.17× narrower
Interval width	1.429 pts	0.451 pts	−0.978 pts
Standard error	0.365 pts	0.115 pts	−0.249 pts
Net result	-241.00 units	-3,672.00 units	-3,431.00 units
Error vs exact	+0.51σ	+0.51σ	—

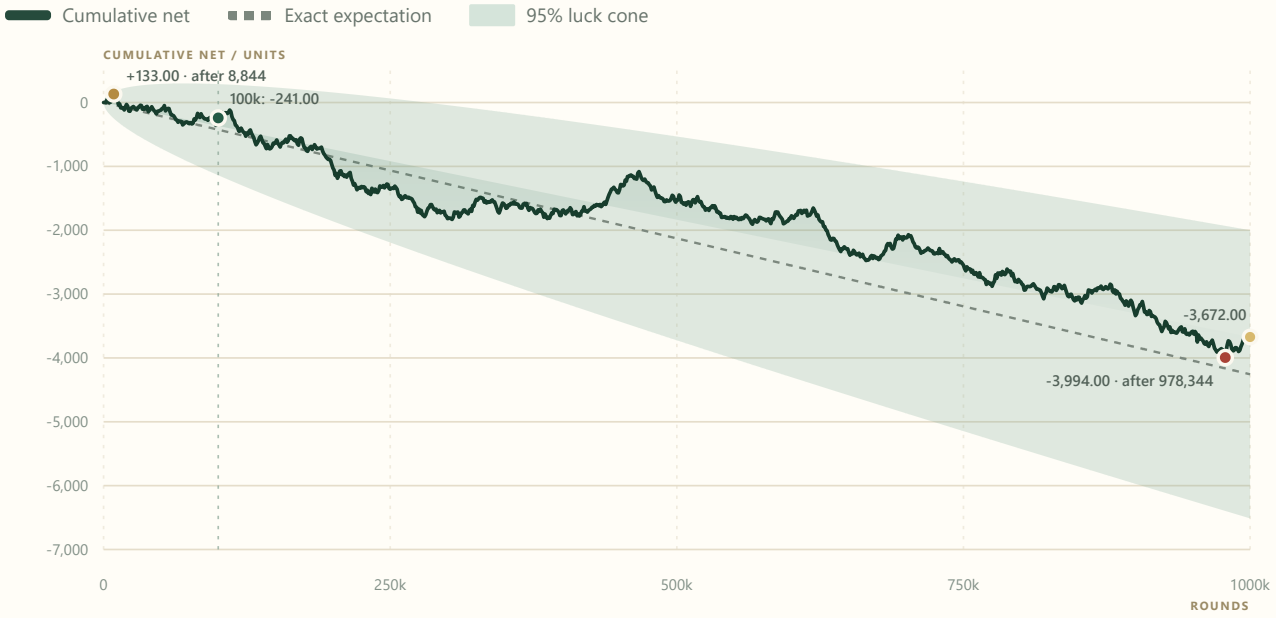
THE SQUARE-ROOT RULE

Ten times the sample does not mean ten times the precision. The standard error scales with **1/√n**, so the interval narrows by **√10 ≈ 3.16×**. That is why the 1M interval is **3.17×** narrower, not 10×.

ONE MILLION ROUNDS

The line keeps *drifting down.*

Cumulative net over 1,000,000 rounds. The marker at 100,000 is the finish of the smaller report; the dashed line is the exact expectation and the cone is the 95% luck range.



Expected path: 0 → -4,256.12. The 95% band at 1,000,000 rounds is roughly ±2,256.96 units.

PEAK
+133.00
after round 8,844

TROUGH
-3,994.00
after round 978,344

MAX DRAWDOWN
4,127.00
units, peak to trough

LONGEST LOSS STREAK
18
rounds in a row

At 100,000 the run was +184.61 units above exact expectation. By 1,000,000 it was +584.12 units above. The sample stayed on the same side of exact as it grew.

The 1M finish is 0.51 standard deviations from exact. The 95% band is ±2,256.96 units at the end. The edge is in the center of the path, not in every step.

WHAT TEN TIMES THE SAMPLE BOUGHT

The observed edge moved from 0.241% to 0.367%. The interval narrowed from 1.429 points to 0.451. The strategy never changed; only the amount of evidence did.

HOW THIS WAS PLAYED

More rounds. *Tighter answer.*

This is the same `PlayGame` table engine, rules and chart policy as the 100k report. Per-decision logging is disabled so a million rounds stay memory-flat.

Rules

Decks	6
Dealer	Stands on soft 17
Blackjack	3:2
Double	Any two · after split
Splits	Up to 4 hands · aces one card
Surrender	None
Dealer peek	US peek
Insurance	Offered; take when EV ≥ 0

Run

Seed	8675309
Rounds	1,000,000
Hands after splits	1,027,805
Bet	1 unit, flat
Penetration	75%
Strategy	Total-dependent chart
First 100k	-241.00 · 43,246–47,547–9,207
Code	6e0fc28

What changed

The 95% interval narrowed from 1.429 points to 0.451 points. The observed edge moved from 0.241% to 0.367%. The 1M interval no longer includes break-even.

This build uses the corrected visible-card insurance density. The card sequence is unchanged; insurance decisions at the threshold now use the pre-hole composition instead of conditioning on the hidden hole.

The first 100,000 rounds are not a new simulation. They are the same sequence as the 100k report: -241.00 units and 43,246–47,547–9,207.

What did not change

The rules, seed, cut card, chart and insurance threshold are identical. The exact edge remains 0.426%. The 1M point estimate sits 0.058 points below it, inside the interval.

Split EV still uses the documented independent-hand approximation. The chart is total-dependent, not composition-dependent.

What this does not prove

The simulation is a consistency check on the engine, not the source of the exact edge. A 1M interval of 0.142% to 0.593% is still ±0.226 points. Getting to ±0.05 points would take roughly **20.4 million rounds**.

REPRODUCE IT

```
node reports/build-blackjack-1m-report.mjs
```

Writes `blackjack-1m.json`, this HTML and the PDF. The seed is 8675309; the 100k report is the first 100,000 rounds of the same sequence.

THE LAST WORD

The edge did not get bigger. The error bar got smaller.